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PostgreSQL: What Really Matters in Configuration, Maintenance, and Performance — On-Prem or in the Cloud

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Introduction

PostgreSQL is a battle-tested open-source database that powers everything from startups to Fortune 100 infrastructure. Whether you're running single PostgreSQL on a bare-metal box in a

operate are **mostly the same** — but how you apply them differs depending on the environment.

This article explores what **really matters** in PostgreSQL configuration, maintenance, and performance tuning — with insights that cut through the noise.

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1. Smart Configuration: The Foundation of Stability

Out-of-the-box, PostgreSQL is configured to run safely on a toaster. If you want real-world performance, **you must tune it**.

Memory Parameters (What You Should Always Tune)

Parameter	Purpose	Rule of Thumb
`shared_buffers`	Caching layer for data	25–40% of total RAM
`work_mem`	Memory for sorting, hashing	4–64MB per session
`maintenance_work_mem`	Used by autovacuum, CREATE INDEX	256MB–2GB
`effective_cache_size`	Planner's view of OS cache	50–75% of RAM

> ***Cloud Tip:*** *On AWS, be conservative with work_mem if you're using many short-lived connections.*

Connection Management

- Set max_connections carefully — higher isn't better
- Use **PgBouncer** to pool connections (especially on cloud-hosted DBs)
- Monitor with pg_stat_activity for runaway sessions

WAL & Replication

- Tune wal_buffers and checkpoint_timeout for better write throughput
- For replication, enable wal_level = logical only when needed
- On cloud: send WAL to object storage (e.g., S3) using tools like wal-g

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Environment-Specific Configuration

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2. Maintenance: The Discipline That Saves You Later

Maintenance is about **predictability**, **data health**, and **preventing future pain**.

Autovacuum: Your Invisible Janitor

PostgreSQL uses **MVCC** (multi-version concurrency control). Without vacuuming, your DB grows forever.

What to Monitor:

```
SELECT relname, n_dead_tup, last_autovacuum FROM pg_stat_user_tables;
```

Best Practices:

- Don't disable autovacuum
- Tune autovacuum_vacuum_threshold and scale_factor based on table size
- Consider manual VACUUM (VERBOSE) for massive bulk loads

Bloat Management

PostgreSQL doesn't overwrite rows — it creates new versions. That's where bloat comes in.

Detect bloat:

Use pgstattuple, pg_bloat_check, or custom queries.

Prevent bloat:

- Use fillfactor
- Avoid frequent updates to wide columns
- Periodically REINDEX heavily used indexes

Backups That Actually Work

It's not backed up until it's **restored and verified**.

On-prem:

- Use pg_basebackup for full logical or physical backups
- Schedule incremental WAL archiving

In the cloud:

- Use pgBackRest, Barman, or wal-g with S3/GCS for tiered backups

Upgrades and Patching

- Use pg_upgrade for major versions
- Always test upgrades in a **replica environment**
- Patch regularly, especially for CVEs and corruption bugs

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3. Performance: Diagnose, Benchmark, Evolve

Performance tuning in PostgreSQL is less about silver bullets and more about **visibility and continuous adjustment**.

Real-World Testing, Not Just Benchmarks

Don't trust pgbench alone. Use:

- Application-level tests
- Production clones
- Monitoring trends over time

The Trinity: Explain, Buffers, Analyze

Use the full EXPLAIN stack:

```
EXPLAIN (ANALYZE, BUFFERS, VERBOSE) SELECT ...;
```

- See which nodes consume I/O, CPU, or memory
- Track sequential scans, index usage, and sort spills

Track the Right Stats

Alerts That Matter

Special Considerations for Cloud Deployments

Summary: What Really Matters

#matters

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